Problem Set 0
Review of Statistics
Econometrics

1. An imaginary country has three autonomous regions. The first one - Rivendell - has two inhabitants who personal yearly income is 30 and 25 thousands of euros. The second region - Rohan - has a population of three with personal yearly income of 45, 62 and 15 thousands of euros. Finally, the third region - Gondor - has five inhabitants with the yearly income of 38, 86, 43, 65 and 24 (thousands of euros).

   (a) Calculate average yearly income for each region.

   (b) Calculate the overall average using the weighted mean (think carefully of the weights).

   Solution:

   (a) The average yearly income for each region:
   For Rivendell:

   $$\text{Average income} = \frac{30+25}{2} = \frac{55}{2} = 27.5 \text{ thousand euros.}$$

   For Rohan:

   $$\text{Average income} = \frac{45+62+15}{3} = \frac{122}{3} = 40.67 \text{ thousand euros (approximately).}$$

   For Gondor:

   $$\text{Average income} = \frac{38+86+43+65+24}{5} = \frac{256}{5} = 51.2 \text{ thousand euros.}$$

   (b) The overall average yearly income using the weighted mean:
   First, note the population of each region:

   - Rivendell: 2 inhabitants
   - Rohan: 3 inhabitants
   - Gondor: 5 inhabitants

   The total population is:

   $$2+3+5=10.$$

   The overall weighted mean is calculated as follows:

   $$\text{Weighted mean} = \frac{(27.5\times 2)+(40.67\times 3)+(51.2\times 5)}{10}.$$

Calculate the numerator:

$$(27.5 \times 2) + (40.67 \times 3) + (51.2 \times 5) = 55 + 122.01 + 256 = 433.01.$$

The weighted mean is:

$$\frac{433.01}{10} = 43.3 \text{ thousand euros.}$$

Thus, the overall average yearly income is 43.3 thousand euros.

2. Company in Madrid has 200 employees and the average monthly salary is 2000 euros with a standard deviation of 1000 euros. The company decided to move to the US and take all the employees with it. Each employee will have his/her salary in euros multiplied by the current EUR/USD exchange rate of 1.2. Additionally, in order to compensate for the increase in healthcare and housing costs, each employee will have a 750 USD increase in their salary. Calculate the new average monthly salary and variance.

   Solution:

   Given:

   - The company has 200 employees.
   - The average monthly salary in euros is $\mu = 2000$ euros.
   - The standard deviation of the salary is $\sigma = 1000$ euros.
   - The EUR/USD exchange rate is 1.2.
   - Each employee will receive a fixed increase of 750 USD.

   We need to calculate the new average monthly salary and the new variance.

   - **Step 1: Calculate the new average monthly salary**
     The linear transformation applied to each employee's salary can be expressed as:

     $$\text{New Salary} = 1.2 \times \text{Salary in Euros} + 750.$$

     The new average salary in USD is:

     $$\text{New Average Salary} = 1.2 \times 2000 + 750 = 2400 + 750 = 3150 \text{ USD.}$$

   - **Step 2: Calculate the new variance**
     For a linear transformation $aX + b$, the variance changes as follows:

     $$\text{New Variance} = a^2 \times \text{Variance of } X.$$

     In this case:

     $$\text{New Variance} = (1.2)^2 \times (1000)^2 = 1.44 \times 1000000 = 1440000 \text{ USD}^2.$$

   - **Conclusion**
     - The new average monthly salary is 3150 USD.

– The new variance is 1440000 USD^2.

3. The following table contains information for 10 days about the average daily temperature in degrees of Celsius (X) and daily electricity consumption in kWh (Y). The last row contains the sums. The table on the right contains some of the descriptive statistics. Use **population** formulas.

X	Y	X^2	Y^2	$X \cdot Y$
23	6	529	36	138
22	5	484	25	110
21	7	441	49	147
24	7	576	49	168
16	6	256	36	96
21	4	441	16	84
18	7	324	49	126
19	6	361	36	114
21	3	441	9	63
17	5	289	25	85
202	56	4142	330	1131

$\bar{x}$	20.20
$\bar{y}$	
σ_x^2	
σ_y^2	1.64
σ_{xy}	-0.02
ρ_{xy}	
a	5.67
x^*	19.00

(a) Fill in the missing values in the Table on the right.

(b) Interpret the sign, strength and significance of the correlation coefficient.

(c) Find the slope of the regression $Y = a + b \cdot X$, interpret.

(d) What is the predicted daily electricity consumption in kWh if the temperature is x^* degrees of Celsius?

Solution:

(a) Fill in the missing values in the table on the right:

- Sample mean $\bar{y}$:

$$\bar{y} = \frac{\sum_{i=1}^{10} Y_i}{10} = \frac{6+5+7+7+6+4+7+6+3+5}{10} = \frac{56}{10} = 5.6.$$

- Population variance σ_x^2:

$$\sigma_x^2 = \frac{\sum_{i=1}^{10}(X_i - \bar{x})^2}{n} = \frac{(23-20.2)^2 + (22-20.2)^2 + \cdots + (17-20.2)^2}{10} = 6.16.$$

- Population correlation coefficient ρ_{xy}:

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \cdot \sigma_y} = \frac{-0.02}{\sqrt{6.16} \cdot \sqrt{1.64}} = -0.006.$$

Hence, the missing values are:

– $\bar{y} = 5.6$

– $\sigma_x^2 = 6.16$

– $\rho_{xy} = -0.006$

(b) Interpret the sign, strength, and significance of the correlation coefficient:

- The correlation coefficient $\rho_{xy} = -0.006$ indicates a very weak linear relationship between temperature X and electricity consumption Y.
- The negative sign suggests a very slight inverse relationship, meaning that as the temperature increases, electricity consumption slightly decreases.
- However, the magnitude is so small that this relationship is practically insignificant, indicating almost no correlation between the variables.

(c) Find the slope of the regression $Y = a + b \cdot X$ and interpret it:

- The slope b is calculated as:

$$b = \frac{\sigma_{xy}}{\sigma_x^2} = \frac{-0.02}{6.16} = -0.0032.$$

- Interpretation: For each additional degree of Celsius in temperature, the electricity consumption Y is expected to decrease by approximately 0.0032 kWh. However, since the slope is very small, this change is negligible.

(d) What is the predicted daily electricity consumption in kWh if the temperature is $x^* = 19$ degrees of Celsius?

- The predicted electricity consumption is calculated using the regression equation:

$$\hat{Y} = a + b \cdot x^*.$$

- First, calculate the intercept a:

$$a = \bar{y} - b \cdot \bar{x} = 5.6 - (-0.0032 \times 20.2) \approx 5.6.$$

- Now, substitute $x^* = 19$ into the regression equation:

$$\hat{Y} = 5.6 + (-0.0032) \times 19 \approx 5.6 \text{ kWh}.$$

- Interpretation: If the temperature is 19 degrees Celsius, the model predicts an electricity consumption of approximately 5.6 kWh.

4. Consider the following regression line that was obtained from a sample of size $n = 80$. Here X is the unemployment rate, and Y is the inflation (N.B. Use sample formulas):

$$Y = 2 - 0.1 \cdot X$$

We know that $\sum y_i = 115$, $\sum y_i^2 = 185$ and $\sum x_i^2 = 2705$.

(a) Calculate the covariance and the coefficient of correlation.

(b) Justify if the variables X and Y are independent or no.

(c) Interpret the coefficients a and b from the economic perspective.

Solution:

(a) Calculate the covariance and the coefficient of correlation using sample formulas:
We are given the following data:

- Sample size $n = 80$
- $\sum y_i = 115$
- $\sum y_i^2 = 185$
- $\sum x_i^2 = 2705$
- The regression equation is $Y = 2 - 0.1 \cdot X$, so the intercept $a = 2$ and the slope $b = -0.1$.

Step 1: Calculate the sample mean of Y:

$$\bar{y} = \frac{\sum y_i}{n} = \frac{115}{80} = 1.4375.$$

Step 2: Calculate the mean of X using the formula for the intercept a:

$$a = \bar{y} - b \cdot \bar{x}$$

Substituting the known values:

$$2 = 1.4375 - (-0.1) \cdot \bar{x}$$

Solving for $\bar{x}$:

$$2 = 1.4375 + 0.1 \cdot \bar{x}$$
$$0.5625 = 0.1 \cdot \bar{x}$$
$$\bar{x} = \frac{0.5625}{0.1} = 5.625.$$

Step 3: Calculate the sample variance of Y (using the sample variance formula):

$$s_y^2 = \frac{\sum y_i^2 - \frac{(\sum y_i)^2}{n}}{n-1} = \frac{185 - \frac{115^2}{80}}{79} \approx \frac{185 - 165.15625}{79} \approx \frac{19.84375}{79} \approx 0.2518.$$

Step 4: Calculate the sample standard deviation of X using the mean $\bar{x} = 5.625$:
The sample variance of X is given by:

$$s_x^2 = \frac{\sum x_i^2 - n \cdot \bar{x}^2}{n-1}.$$

Substituting the values:

$$s_x^2 = \frac{2705 - 80 \cdot (5.625)^2}{79} = \frac{2705 - 80 \cdot 31.640625}{79} = \frac{2705 - 2531.25}{79} = \frac{173.75}{79} \approx 2.1994.$$

Taking the square root:

$$s_x = \sqrt{2.1994} \approx 1.483.$$

Step 5: Calculate the sample covariance s_{xy}: The sample covariance for the regression equation $Y = a + b \cdot X$ is given by:

$$s_{xy} = b \cdot s_x^2 = -0.1 \times 2.1994 = -0.2199.$$

Step 6: Calculate the sample correlation coefficient r_{xy}: The sample correlation coefficient is given by:

$$r_{xy} = \frac{s_{xy}}{s_x \cdot s_y} = \frac{-0.2199}{1.483 \times \sqrt{0.2518}} \approx -0.282.$$

So, the sample covariance is -0.2199 and the sample correlation coefficient is -0.282.

(b) Justify if the variables X and Y are independent:

The sample correlation coefficient $r_{xy} = -0.282$ indicates a weak negative linear relationship. However, **correlation alone cannot be used to claim independence**. Independence would require all joint distributions to be considered, not just linear relationships.

(c) Interpret the coefficients a and b from an economic perspective:

- The intercept $a = 2$ represents the predicted inflation rate when the unemployment rate is zero. In economic terms, even with no unemployment, there is still some expected inflation (2
- The slope $b = -0.1$ indicates that for each 1% increase in the unemployment rate, inflation is expected to decrease by 0.1%. This is consistent with the inverse relationship described by the Phillips curve, which suggests that higher unemployment typically leads to lower inflation.

5. The value of an investment portfolio follows a Normal distribution with a mean value of €500,000 and a standard deviation of €15,000. Calculate the probability that the value of your portfolio is between €485,000 and €530,000.

Solution:

Since the value of the portfolio is normally distributed with mean $\mu = 500,000$ and standard deviation $\sigma = 15,000$, we can calculate the desired probability by finding the area under the normal curve between the two values of $485,000$ and $530,000$. To do this, we first need to standardize the interval by subtracting the mean and dividing by the standard deviation. Let z_1 and z_2 be the standardized values for the lower and upper bounds, respectively. Then,

$$z_1 = \frac{485,000 - 500,000}{15,000} = -1,$$
$$z_2 = \frac{530,000 - 500,000}{15,000} = 2.$$

The desired probability is then given by the area under the standard normal curve between z_1 and z_2, which can be found using a standard normal table or using software such as R.

Using a standard normal table, we find that $F(-1) = 0.1587$ and $F(2) = 0.9772$, so the desired probability is $0.9772 - 0.1587 = 0.8185$.

Therefore, the probability that the value of the portfolio is between €485,000 and €530,000 is 0.8185.

Here, we will use R to calculate the probability:

```
z1 <- -1
z2 <- 2
pnorm(z2) - pnorm(z1)
```

6. Suppose that a random variable X follows a normal distribution with mean 66 and variance 100. Calculate the value of k such that $P(60 < X < k) = 0.2496$.

Solution:

Given that a random variable X follows a normal distribution with mean $\mu = 66$ and variance $\sigma^2 = 100$, we need to calculate the value of k such that $P(60 < X < k) = 0.2496$.

First, we convert the given probabilities into the standard normal distribution using the Z-score formula:

$$Z = \frac{X - \mu}{\sigma}$$

For $X = 60$, the Z-score is:

$$Z_{60} = \frac{60 - 66}{10} = -0.6$$

The probability of X being less than 60 is given by the cumulative distribution function (CDF) of the standard normal distribution at Z_{60}:

$$P(X < 60) = P(Z < -0.6)$$

Given that $P(60 < X < k) = 0.2496$, we need to find the cumulative probability $P(X < k)$ that includes both $P(X < 60)$ and $P(60 < X < k)$:

$$P(X < k) = P(X < 60) + P(60 < X < k)$$

Using the CDF of the standard normal distribution, we solve for k:

$$P(X < k) = P(Z < -0.6) + 0.2496$$

Let Z_k be the Z-score corresponding to k. Then:

$$P(Z < Z_k) = P(X < k)$$

Solving for k using the CDF of the standard normal distribution, we find:

$$k = \sigma \cdot Z_k + \mu$$

Substituting the values:

$$k \approx 10 \cdot Z_k + 66$$

Given the calculated cumulative probability up to k, we find $k \approx 66.60$.

7. Annual percentage salary increases for managers in medium-sized companies follow a normal distribution with a mean of 12.2% and a standard deviation of 3.6%. A random sample of nine observations is drawn from this population, and the sample mean is calculated. What is the probability that the sample mean is less than 10%?

Solution:

We are given that the distribution of annual percentage salary increases for managers follows a normal distribution with a mean of 12.2% and a standard deviation of 3.6%. We are also told that a random sample of 9 observations is drawn from this population, and we need to find the probability that the sample mean is less than 10%.

The sample mean, denoted by $\bar{X}$, is a random variable itself and follows a normal distribution with mean equal to the population mean, 12.2%, and standard deviation equal to the population

standard deviation divided by the square root of the sample size, which is 9 in this case. Thus, we have:

$$\bar{X} \sim N\left(\mu = 12.2\%,\ \sigma = \frac{3.6\%}{\sqrt{9}} = 1.2\%\right)$$

We want to find the probability that the sample mean is less than 10%, which is equivalent to finding the probability that a standard normal variable, Z, is less than:

$$Z = \frac{\bar{X} - \mu}{\sigma} = \frac{10\% - 12.2\%}{1.2\%} = -1.83$$

Using a standard normal distribution table or calculator, we can find that $P(Z < -1.83) \approx 0.0336$. Therefore, the probability that the sample mean is less than 10% is approximately 0.0336 or about 3.36%.

8. Exercise 15. One year the percentage rates of return of mutual funds followed a normal distribution with a mean of 14.8 and a standard deviation of 6.3 . A random sample of nine of these funds was taken.

   - What is the probability that the sample mean of the percentage rates of return is greater than 19 ?
   - What is the probability that the sample mean of the percentage rates of return is between 10.6 and 19 ?
   - The probability that the sample mean percentage rate of return is less than is 0.2483 ?

   Solution:

   One year the percentage rates of return of mutual funds followed a normal distribution with a mean of 14.8 and a standard deviation of 6.3. A random sample of nine of these funds was taken.

   a) What is the probability that the sample mean of the percentage rates of return is greater than 19?

$$P(\bar{x} > 19) = P(\frac{\bar{x} - \mu}{\sigma_{\bar{x}}} > \underbrace{\frac{19 - \mu}{\sigma_{\bar{x}}}}_{\frac{\sigma}{\sqrt{n}}})$$

$$= P\left(z > \frac{19 - 14.8}{\frac{6.3}{\sqrt{9}}}\right) = P(z > 2) = 0.0228 = 2.28\%$$

   b) What is the probability that the sample mean of the percentage rates of return is between 10.6 and 19?

$$P(10.6 < \bar{x} < 19) = P\left(\frac{10.6-\mu}{\underbrace{\sigma_{\bar{x}}}_{\frac{\sigma}{\sqrt{n}}}} < \frac{\bar{x}-\mu}{\sigma_{\bar{x}}} < \frac{19-\mu}{\underbrace{\sigma_{\bar{x}}}_{\frac{\sigma}{\sqrt{n}}}}\right)$$
$$= P\left(\frac{10.6-14.8}{\frac{6.3}{\sqrt{9}}} < z < \frac{19-14.8}{\frac{6.3}{\sqrt{9}}}\right) = P(-2 < z < 2)$$
$$= P(z < 2) - P(-2 < z) = P(z < 2) - 1 + P(z < 2) = 0.9544 = 95.44\%$$

c) The probability that the sample mean percentage rate of return is less than _____ is 0.2483?

We want to find a such that $P(\bar{x} < a) = 0.2483$. Therefore,

$$P(\underbrace{\frac{\bar{x}-\mu}{\sigma_{\bar{x}}}}_{z} < \underbrace{\frac{a-\mu}{\sigma_{\bar{x}}}}_{z_a}) = 0.2483$$

Notice that the value of z_a has to be negative. This is because when $P = 0.5$, then $z = 0$. Because $0.2483 < 0.5$ it follows that $z_a < 0$. Therefore,

$$P\left(z < \frac{a-\mu}{\sigma_{\bar{x}}}\right) = 1 - 0.2483 = 0.7517 \rightarrow z = 0.68 \rightarrow z_a = -0.68 = \frac{a-\mu}{\sigma_{\bar{x}}} = \frac{a-14.8}{\frac{6.3}{\sqrt{9}}} \rightarrow a = 13.372$$

9. Given a population $\mathrm{N}(\mu, \sigma = 7)$ and the estimators of the population mean μ, for simple random samples of size n = 3

$$\hat{\theta}_1 = \frac{1}{3}(X_1 + X_2 + X_3)$$
$$.\hat{\theta}_2 = \frac{1}{2}(X_1 + \frac{1}{3}X_2 + \frac{1}{4}X_3)$$

a) Check that the estimators $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are or are not unbiased.
b) Calculate the variance of both estimators. Which one is more efficient?

Solution:

Given a population $N(\mu, \sigma = 7)$ and the estimators of the population mean μ, for simple random samples of size $n = 3$

$$\hat{\theta}_1 = \frac{1}{3}(X_1 + X_2 + X_3)$$
$$\hat{\theta}_2 = \frac{1}{2}X_1 + \frac{1}{3}X_2 + \frac{1}{4}X_3$$

a) Check that the estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased or not;

b) Calculate the variance of both estimators. Which is more efficient? Relative efficiency?

Solution

a)

$$E\left[\hat{\theta}_1\right] = E\left[\frac{1}{3}\left(X_1 + X_2 + X_3\right)\right] = \frac{1}{3}(3\mu) = \mu \rightarrow \text{ Unbiased}$$
$$E\left[\hat{\theta}_2\right] = E\left[\frac{1}{2}X_1 + \frac{1}{3}X_2 + \frac{1}{4}X_3\right] = \frac{1}{2}\mu + \frac{1}{3}\mu + \frac{1}{4}\mu = \frac{13}{12}\mu \rightarrow \text{Biased}$$

b)

$$\text{Var}\left(\hat{\theta}_1\right) = \text{Var}\left(\frac{1}{3}\left(X_1 + X_2 + X_3\right)\right) = \frac{1}{9}\left(3\sigma^2\right) = \frac{49}{3}$$
$$\text{Var}\left(\hat{\theta}_2\right) = \text{Var}\left(\frac{1}{2}X_1 + \frac{1}{3}X_2 + \frac{1}{4}X_3\right) = \frac{1}{4}\sigma^2 + \frac{1}{9}\sigma^2 + \frac{1}{16}\sigma^2 = \frac{2989}{144}$$

We observe that $\hat{\theta}_1$ has a lower variance in comparison to $\hat{\theta}_2$, therefore it is more efficient.

10. In a population $N(\mu; 2)$ the value of the mean is unknown, establishing two hypotheses about it: null hypothesis $\mu_0 = 4$ and alternative hypothesis $\mu_1 = 6$. We will calculate the critical region with a significance level $\alpha = 0.05$, using a simple random sample of size 4, whose elements are: $1.8; 6.6; 2.4; 3.5$.
a) Determining whether the null hypothesis is accepted or rejected.
b) Carry out the exercise for when the null hypothesis is $\mu_0 = 6$ and the alternative hypothesis is $\mu_1 = 4$

Solution:

a) The sample mean is $\bar{x} = 3.575$. The test statistic is:

$$\begin{aligned} Z &= \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \\ &= \frac{3.575 - 4}{2/\sqrt{4}} \\ &= -0.425 \end{aligned}$$

Since the null hypothesis is $\mu = 4$ and the alternative hypothesis is $\mu = 6$, we are testing whether the sample mean is significantly greater than the null hypothesis mean. In this case, we should be looking at the upper tail of the distribution.

For a one-tailed test with a significance level $\alpha = 0.05$, we should find the Z-score corresponding to the $1 - \alpha = 0.95$. That Z-score is 1.645.

Since our test statistic is smaller than the critical value $Z_{(1-\alpha)} = 1.645$, we fail to reject the null hypothesis.

In this case, there is not enough evidence to support the alternative hypothesis (H_1) that $\mu = 6$.

b) Since the null hypothesis is $\mu = 6$ and the alternative hypothesis is $\mu = 4$, we are testing whether the sample mean is significantly lower than the null hypothesis mean. In this case, we should be looking at the lower tail of the distribution.

For a one-tailed test with a significance level $\alpha = 0.05$, the Z-score is -1.645.

Since our test statistic

$$\begin{aligned} Z &= \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \\ &= \frac{3.575 - 6}{2/\sqrt{4}} \\ &= -2.425 \end{aligned}$$

is below the critical value $Z_{(\alpha)} = -1.645$, we reject the null hypothesis.

In this case, there is evidence to support the null hypothesis (H_0) that $\mu = 6$.

11. To test the null hypothesis $H_0[\mu = 3]$ against the alternative hypothesis $H_1[\mu \neq 3]$ in the normal distribution $N(\mu, \sigma^2)$, a significance level of 0.013 is used. With the population variance unknown, a simple random sample of size 18 is taken. The sample mean is $\bar{x} = 3.911$, and the sum of squared deviations from the sample mean is $\sum_{i=1}^{18}(x_i - 3.911)^2 = 0.762698$. Determine whether to accept or reject the null hypothesis based on this sample data.

Solution:

To test the null hypothesis $H_0[\mu = 3]$ against the alternative hypothesis $H_1[\mu \neq 3]$, we will use a two-tailed t-test, assuming that the population variance is unknown. We are given the sample size, sample mean, the sum of squared deviations from the sample mean, and the significance level.

Sample size: $n = 18$

Sample mean: $\bar{x} = 3.911$

Sum of squared deviations: $\sum_{i=1}^{18}(x_i - 3.911)^2 = 0.762698$

Significance level: $\alpha = 0.013$

First, we calculate the sample standard deviation:

$$s = \sqrt{\frac{\sum_{i=1}^{18}(x_i - 3.911)^2}{n-1}} = \sqrt{\frac{0.762698}{17}} \approx 0.212$$

Next, we compute the t-score:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{3.911 - 3}{\frac{0.212}{\sqrt{18}}} \approx 18.247$$

Note that, since we do not know the population variance, we should use a t-distribution for calculating the t-score. Now, we find the critical values for a two-tailed t-test with 17 degrees of freedom and a significance level of 0.013. Using a t-distribution table or a t-distribution calculator, we find the critical values to be approximately ± 2.773. Since our t-score is greater than the critical value ($18.247 > 2.773$), we reject the null hypothesis $H_0[\mu = 3]$. Based on the sample data, we have enough evidence to support the alternative hypothesis $H_1[\mu \neq 3]$.

12. The advertising of an investment fund states that its average annual return is greater than 7%. To support this claim, the publicity indicates that the annual returns of the fund in the last 5 years were $8\%, 6.1\%, 9\%, 12\%$, and 10%.
a) Assuming that the annual returns of the fund are independent and identically distributed with a normal distribution, propose the appropriate hypothesis test to test the advertising statement, identifying the null and alternative hypotheses.
b) Calculate the test statistic. Solve the test with a significance level of 5%, and interpret the result.
c) Approximately calculate (bound) the p-value. Indicate for which significance levels you can affirm that the null hypothesis is not rejected with the data obtained. Interpret the result.
d) Now suppose that the standard deviation of the fund's annual return is known and equal to 0.022. Under this assumption, solve section b again and comment on the result ($\alpha = 5\%$).
e) Under this last hypothesis, calculate the power of the test for an average annual return of 9% and a significance level of 5%. Interpret the result.

Solution:

a) We are testing whether the average annual return of the fund is greater than 7%. The null hypothesis H_0 is that the average annual return is less or equal to 7% ($H_0 : \mu \leq 0.07$), and the alternative hypothesis H_1 is that the average annual return is greater than 7% ($H_1 : \mu > 0.07$).

b) First, we calculate the sample mean and sample standard deviation. The sample mean is

$$\bar{x} = \frac{1}{5}(0.08 + 0.061 + 0.09 + 0.12 + 0.10) = 0.0902$$

The sample standard deviation is

$$s = \sqrt{\frac{1}{4}\left[(0.08 - 0.0922)^2 + (0.061 - 0.0922)^2 + \ldots + (0.10 - 0.0922)^2\right]} \approx 0.022$$

The sample size is $n = 5$.

The test statistic is a t-score, given by

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $\mu_0 = 0.07$ is the hypothesized mean. We have

$$t = \frac{0.0902 - 0.07}{0.022/\sqrt{5}} \approx 2.05$$

With a significance level of 5% and $n - 1 = 4$ degrees of freedom, the critical value $t_{0.05,4}$ is approximately 2.132. Since the test statistic is lower than the critical value, we do not reject the null hypothesis.

c) The p-value is the probability of obtaining a test statistic as extreme as, or more extreme than, the observed test statistic under the null hypothesis. Since this is a one-tailed test, the p-value is $P(T > t)$, where T is a random variable following a t-distribution with $n - 1 = 4$ degrees of freedom. To bound the p-value, we can use the fact that $P(T > 2.776) = 0.05$ and $P(T > 1.533) = 0.10$. Since our observed test statistic is $1.533 < 2.05 < 2.776$, we can safely say that $0.05 <$ p-value < 0.10.

d) If the standard deviation is known and equal to 0.022, we use a z-test instead of a t-test. The test statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{0.0902 - 0.07}{0.022/\sqrt{5}} \approx 2.05$$

With a significance level of 5%, the critical value $z_{0.05}$ is approximately 1.645. Since the test statistic is greater than the critical value, we reject the null hypothesis.

e) The power of the test is the probability of correctly rejecting the null hypothesis when the true average annual return is 9%. It can be calculated as $1 - \beta$, where β is the probability of Type II error or, equivalently, the probability of *not* rejecting the null hypothesis when the true average annual return is 9%.

First, let's write down the decision rule assuming that the standard deviation is known and equal to 0.022:

$$\text{reject } H_0 \text{ if } \frac{\bar{x} - 0.07}{0.022/\sqrt{5}} > 1.645 \text{ or } \bar{x} > 0.07 + 1.645(0.022/\sqrt{5}) = 0.086.$$

The probability of Type II error when the true average annual return is 9% is calculated as follows:

$$\begin{aligned}\beta = P(\bar{x} \leq 0.086 \mid \mu = 0.09) &= P\left(z \leq \frac{0.086 - 0.09}{0.022/\sqrt{5}}\right) \\ &= P(z \leq -0.406) \\ &= 0.367\end{aligned}$$

The probability, β, of Type II error when the population mean is 9% is 0.367. Since the power of a test is 1 minus the probability of Type II error, when the population mean is 9%, we have the following:

$$\text{power } = 1 - \beta = 1 - 0.367 = 0.633.$$